INTERIM FEDERAL SPECIFICATION

CLEANING COMPOUND, ENGINE COOLING SYSTEM

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, D. C. 20407, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. This specification covers a cleaning compound and neutralizer for cooling systems of engines on motor vehicles.

1.2 Classification. The cleaning compound and neutralizer covered by this specification shall be of one grade.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS.

2.1 Specifications and Standards. The following specifications and standards, of the issue in effect on the date of invitation for bids, form a part of this specification:

Federal Specifications:

O-O-690 - Oxalic Acid, Dihydrate, Technical
PPP-B-636 - Box, Fiberboard

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels
Fed. Std. No. 123 - Marking for domestic shipment (civilian agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6850

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2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Federal Hazardous Substance Labeling Act.

(Copies may be obtained from the U. S. Department of Health, Education and Welfare, Food and Drug Administration, Washington, D. C. 20407.)

American Society for Testing and Materials.

ASTM D 501 - 58 - Sampling and Chemical Analysis of Alkaline Detergents

ASTM D 1331-56 - Test for Surface and Interfacial Tension of Solutions of Surface-Active Agents.

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

3. REQUIREMENTS.

3.1 Components. The cleaner shall consist of a major component and a minor component.

3.1.1 Major component. The major component shall consist of oxalic acid and a wetting agent. The major component shall consist of not less than 98-percent oxalic when tested as specified in 4.4.1.

3.1.2 Minor component. The minor component shall consist of sodium carbonate and a wetting agent. The minor component shall consist of not less than 98-percent anhydrous sodium carbonate in powder or granular form and shall meet the test specified in 4.4.2.

3.2 Surface tension. The major and the minor component, when tested as specified in 4.4.3, shall each show a surface tension of not more than 37 dynes per centimeter at 25°C.

3.3 Practical test. The major component shall dissolve not less than 30 mg. of rust when tested as specified in 4.4.4.

3.4 Material. Both the major component and the minor component shall be free flowing, free of dirt and foreign material when tested as specified in 4.4.5.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection. Sampling for inspection shall be in accordance with provisions of MIL-STD-105, except where otherwise indicated.

4.2.1 Inspection of materials and components. The supplier is responsible for insuring that materials and components used were tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification.

4.3 Inspection of the end item.

4.3.1 Examination of the end item for defects in packaging and marking. The sample unit for this examination shall be one filled shipping container.

Examine	Defect
Containers	Shipping container not as specified.
Marking	Omitted, illegible, incorrect, or incomplete.
Instruction label	Omitted, illegible, incorrect, or incomplete.
Unit container	Minor component not in bottom compartment.

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4.3.2 Examination of the end item. Each unit of the sample shall be inspected for compliance with 3.4.

4.3.3 Examination of filled containers to determine compliance for contents. The average weights of the major and minor components in the sample shall not be less than specified.

4.3.4 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels, for determining the sample size, and the acceptable quality levels (AQLs), expressed in percent defective, shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.1	S-4	4.0
4.3.2	S-2	2.5
4.3.3	S-2	---

4.3.5 Testing of the end item. The sample size shall be five. The test sample unit shall be a 16-ounce composite of the major component and 5-ounce composite of the minor component of the five sample specimens.

4.4 Test procedures.

4.4.1 The major component when tested by the sodium hydroxide titration and the potassium permanganate method of O-O-690, shall meet the requirement of 3.1.1.

4.4.2 The minor component shall be tested in accordance with ASTM Method D 501-58, Sections 19, 20, and 21 to determine compliance with 3.1.2.

4.4.3 Surface tension. One part of the major component compound dissolved in 40 parts of distilled water shall be tested at 25°C in accordance with ASTM Method D 1331-56 to determine compliance with 3.2. The minor component compound shall be tested in like manner to determine compliance with 3.2.

4.4.4 Practical test.

4.4.4.1 Preparation of rust paste. Six clean cast iron discs (2" diameter x 1/4" thick made from GC Meehanite) separated by Z-shaped glass rods are placed horizontally at the bottom of a 600 ml. high-form lipless beaker. Approximately 300 ml. of distilled water containing 0.05% acetic acid are added and the beaker closed with a rubber stopper fitted with an aeration tube (6 mm. I.D. plain tube) and a water-cooled reflux condenser. The solution is heated at 170 degrees F. with an aeration rate of 0.028 cfm (790 ml/min) for 72 hours. The freshly prepared rust is filtered off and oven dried at 220 degrees F. A paste is then made by thoroughly mixing 80% by weight of the rust with 20% Tycol Amulkut #4 (Tidewater Oil Company or equal). One beaker should provide sufficient rust for one test evaluation.

4.4.4.2 Procedure. Dissolve one full can of the major component (16 ounces) in sufficient distilled water to make a 3.0% solution. Weigh out 4 grams of rust paste on a watch glass and add to a beaker containing 350 ml. of the 3% cleaner solution, heated to 170°F. Maintain this temperature with constant stirring for one hour. After one hour pipette out 10 ml. of the suspension, filter into a weighed crucible and ash the precipitate.
Reweigh and determine the weight of dissolved rust.

4.4.5 Material. Both the major and minor components shall be visually inspected for compliance with 3.4.

5. PREPARATION FOR DELIVERY.

(Fed. Std. No. 102 shall be referred to for definitions and applications of the various levels of packaging and packing for civil agency procurement.)

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5.1 Packaging.

5.1.1 Levels A, B and C. Unless otherwise specified in the contract or order, the components shall be packaged in a standard commercial type two compartment paperboard can with slip-on lids and metal ends. The lids shall be adequately sealed and shall be provided with a quick opening device for each compartment. Each container shall hold a minimum of 16-ounces of the major component and 2 ounces of the minor component.

5.2 Packing. Packing shall be level A, B or C as specified (6.2).

5.2.1 Level A. Twelve containers, packaged as specified in 5.1, shall be packed in a snug-fitting, fiberboard shipping container conforming to V3c or V3s of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification.

5.2.2 Level B. Twelve containers, packaged as specified in 5.1, shall be packed in a snug-fitting, fiberboard shipping container conforming to PPP-B-636, type CF or SF, class-domestic. Closure shall be in accordance with the appendix to the box specification.

5.2.3 Level C. The compound, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Labeling.

5.4.1 Instructions for use. Each container shall have imprinted on the label in waterproof ink, the following instructions:

Drain the cooling system completely, then close drain cocks. Pour entire contents of upper compartment into the radiator. (Use two cans for systems holding more than 22 quarts). Fill system with water and replace radiator cap. Avoid overfilling and spillage of solution on car finish. Run engine at fast idle and warm up to 180-190 degrees F. Continue to run for thirty minutes to one hour, maintaining this temperature. If necessary, partly cover radiator to avoid boiling. (Very dirty systems may require up to two hours running or an extra cleaning operation.) Stop engine and drain solution completely from radiator, engine block, and heater. Close drain opening and refill with water. Pour neutralizer from bottom compartment of the container into the radiator, circulate, and drain. Close all cocks and fill system for running.

5.4.2 The container label shallbe marked to show where container is to be opened for both the cleaner and the neutralizer.

5.4.3 Warning statements. In addition to complying with the requirements of the Federal Hazardous Substances Labeling Act, each can shall have imprinted with adequate contrast, Not less than the following:

POISON (18 point type minimum)
MAY BE FATAL IF SWALLOWED (12 point type minimum)
(The above must be on front panel)
Contains oxalic acid - avoid contact with eyes or prolonged contact with skin.

ANTIDOTE: Eyes: flush thoroughly with water, call physician.
Internal: Give large quantities of milk or chalk and induce vomiting (tablespoon of salt in glass of warm water)
Call physician immediately
Keep out of reach of children

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6. NOTES.

6.1 Intended use. The cleaning compound covered by this specification is intended for cleaning water cooling systems of engines on motor vehicles. It is a two step type cleaner, consisting of an oxalic acid type cleaner to assist in dissolving and removing accumulated rust and dirt. The engine must then be flushed with a neutralizer compound. Normal maintenance procedure after the cleaning operation would be to refill the cooling system with an antifreeze containing a corrosion inhibitor or with water to which has been added a corrosion inhibitor conforming to O-I-490 (6840-584-2707).

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Specify quantity required.
(c) Level of packaging and packing (5.1 and 5.2)

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